



# **NATIONAL UNIVERSITY OF MODERN LANGUAGES**

## **Computer Networks lab Report - 8**

**NAME: Abdul Rafay**

**ROLL NO: FL23791**

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**SUBMITTED TO: Mam Anam Taskeen**

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# Switch Redundant Connections and Loops (Spanning Tree Protocol – STP)

## Objectives

The objectives of this lab are:

- To understand the concept of **Spanning Tree Protocol (STP)**.
- To study how STP prevents loops in redundant switch networks.
- To configure a **Root Bridge** manually.
- To enable **Rapid Spanning Tree Protocol (RSTP)**.
- To analyze **root, forwarding, and blocked ports** in a switched network.

## Introduction

In modern network topologies, switches are often connected using **redundant links** to improve reliability and avoid a single point of failure. While redundancy increases availability, it also introduces **network loops**, which can cause broadcast storms and network failure.

To overcome this problem, **Spanning Tree Protocol (STP)** is used. STP logically disables redundant links and creates a **loop-free tree topology**, while still keeping backup paths available.

At the center of the STP network is the **Root Bridge**, which is the most important switch. All other switches calculate the best path to reach the Root Bridge and block unnecessary links.

## Root Bridge Selection

The Root Bridge is selected based on the **Bridge ID**, which consists of:

- **Priority number**
- **MAC address**

By default:

- All switches have the same priority value (**32768**)
- The switch with the **lowest MAC address** becomes the Root Bridge

Since MAC addresses cannot be changed, we influence root selection by changing the **priority value**.

Lower priority = higher chance of becoming Root Bridge.

Priority values:

- Range: 0 to 61440

- Must be a **multiple of 4096**

### **Rapid Spanning Tree Protocol (RSTP)**

RSTP is an improved version of STP that provides:

- Faster convergence
- No dependency on long timers
- Rapid recovery from topology changes

RSTP supports a maximum network diameter of **40 switches** and is commonly used in modern networks.

### **Lab Tasks**

#### **Task 1: Configure Switch 1 as Root Bridge**

To make **Switch 1** the Root Bridge for **VLAN 1**, the following command is used:

```
Switch(config)# spanning-tree vlan 1 priority 0
```

**OR**

```
Switch(config)# spanning-tree vlan 1 root primary
```

This sets the lowest priority, ensuring Switch 1 becomes the Root Bridge.

#### **Verification of Root Bridge**

To verify the Root Bridge and port roles, the following command is used:

```
Switch# show spanning-tree
```

This command displays:

- Root Bridge information
- Port states (Root, Designated, Forwarding, Blocked)
- Path cost

#### **Subtask: Switch Port Configuration**

All switch ports are configured as **trunk ports**.

#### **Port Configuration Requirements**

- Ports: FastEthernet 0/1 – 0/5
- Speed: 100 Mbps
- Duplex: Full

- VLAN: 1
- Network: 192.168.1.0
- VLAN state: No shutdown

## **Task 2: Enable Rapid Spanning Tree Protocol (RSTP)**

RSTP is enabled on **all switches** using the following command:

```
Switch(config)# spanning-tree mode rapid-pvst
```

This enables faster convergence and improved performance.

### **Checking Port States**

To identify:

- Root Ports
- Forwarding Ports
- Blocked Ports

The following command is used:

```
Switch# show spanning-tree
```

Blocked ports confirm that STP is successfully preventing loops.

### **Result**

- Switch 1 was successfully selected as the **Root Bridge**.
- Redundant links were automatically blocked to prevent loops.
- Rapid Spanning Tree Protocol was enabled on all switches.
- Network remained loop-free while maintaining redundancy.